MF-02: A uniform constant-factor bound for cubic composition overhead

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This manuscript establishes a uniform constant-factor asymptotic bound for the stage count in MF-02. The classical cubic iteration and the prior constant-factor optimality results are credited in Section 6. The explicit estimate proved here is at most one cubic stage per allowed unrestricted multiplication. The exact smallest stage count and the stronger same-budget optimization question remain open.

Prepared with substantial ChatGPT/Codex assistance at the author's request. A separate independent mathematical review returned PASS for the theorem and the canonical uniform asymptotic-order scope. A second independent source and scope review confirms the attribution and the distinction between the canonical asymptotic target and the stronger source questions. This is an informal automated-agent audit, not external human peer review or formal verification.

1. Exact target and result

Let $I_\delta = [-1,-\delta] \cup [\delta, 1]$, where $0 < \delta < 1$. Let $\mathcal{P}_m$ be the real polynomials computed from $1, x$, using arbitrary real linear combinations and at most m nonscalar multiplications. Define

$$E_m(\delta) = \inf_{p \in \mathcal{P}_m} \max_{x \in I_\delta} |p(x) - \operatorname{sign}(x)|.$$

For $T \geq 0$, let

$$C_T(\delta) = \inf_{a_t, b_t \in \mathbb{R}} \max_{x \in I_\delta} |(q_T \circ \cdots \circ q_1)(x) - \operatorname{sign}(x)|, \qquad q_t(x) = a_t x + b_t x^3.$$

At $T = 0$ the composition is x. The canonical MF-02 target asks for the asymptotic dependence on both m, δ of

$$T_{\min}(m, \delta) = \inf\{T \in \mathbb{N}_0 : C_T(\delta) \leq E_m(\delta)\}.$$

Theorem. For every $0 < \delta < 1$,

$$T_{\min}(0, \delta) = T_{\min}(1, \delta) = 1.$$

For every integer $m \geq 2$,

$$\left\lfloor \frac{m}{2} \right\rfloor \leq T_{\min}(m, \delta) \leq m. \tag{1}$$

Consequently,

$$\frac{m+1}{4} \leq T_{\min}(m, \delta) \leq m + 1 \qquad (m \in \mathbb{N}_0,\ 0 < \delta < 1). \tag{2}$$

Thus the asymptotic order is $\Theta(m+1)$, with absolute constants uniform over the entire gap range, including gaps depending on m.

This theorem gives a constant-factor asymptotic comparison. It does not give an exact formula for $T_{\min}$, an optimal leading constant, or the value of E_m. The uniform two-sided order in (2) determines the canonical asymptotic dependence on both parameters. The broader source question about the exact smallest T, and the comparison at the same multiplication budget, remain unanswered by (1).

2. A degree-only lower bound

Put

$$r_\delta = \frac{1-\delta}{1+\delta} \in (0,1).$$

Lemma 1. If $p \in \mathbb{R}[x]$ has degree at most D, where $D \geq 1$ is an integer, then

$$\max_{x\in I_\delta} |p(x) - \operatorname{sign}(x)| \geq r_\delta^D. \tag{3}$$

We first recall and prove the needed Chebyshev comparison. Let T_D be the degree-D Chebyshev polynomial, characterized by

$$T_D(\cos\theta) = \cos(D\theta).$$

If R is a real polynomial of degree at most D with $|R| \leq 1$ on $[-1,1]$, then

$$|R(t)| \leq |T_D(t)| \qquad (|t| > 1). \tag{4}$$

For $t_0 > 1$, suppose otherwise. Replacing R by $-R$ if necessary, set $c = R(t_0)/T_D(t_0) > 1$. At the $D+1$ points $\cos(j\pi/D)$, the polynomial $R - cT_D$ has strict alternating signs, because $|R| \leq 1 < c$. It has at least D distinct roots in $(-1,1)$, as well as the root t_0, contradicting its degree. The polynomial is not identically zero because $c > 1$. The case $t_0 < -1$ follows by replacing $R(t)$ with $R(-t)$ and using $T_D(-t) = (-1)^D T_D(t)$.

To prove (3), let the error of p be e. If $e \geq 1$, the assertion follows. The value $e = 0$ is impossible: agreement with 1 throughout the interval $[\delta, 1]$ would force $p \equiv 1$, contradicting agreement with -1 on the other interval. Hence consider $0 < e < 1$. The odd part

$$o(x) = \frac{p(x) - p(-x)}{2}$$

also has error at most e, so $1 - e \leq o(x) \leq 1 + e$ on $[\delta, 1]$. As o is odd, $o(x)^2 = B(x^2)$ for a real polynomial B of degree at most D, and $B(0) = 0$. The polynomial

$$H(t) = \frac{1 + e^2 - B(t)}{2e}$$

has absolute value at most 1 on $[\delta^2, 1]$. Map this interval affinely to $[-1,1]$; the image of $t = 0$ is

$$-\frac{1+\delta^2}{1-\delta^2}.$$

By (4),

$$\frac{1+e^2}{2e} = H(0) \leq T_D\left(\frac{1+\delta^2}{1-\delta^2}\right) = \frac{r_\delta^{-D} + r_\delta^D}{2}. \tag{5}$$

The last equality follows from $T_D(\cosh s) = \cosh(Ds)$ and

$$\frac{1+\delta^2}{1-\delta^2} = \cosh\left(\log\frac{1+\delta}{1-\delta}\right).$$

The function $e \mapsto (e + e^{-1})/2$ is strictly decreasing on $(0,1)$. Thus (5) gives $e \geq r_\delta^D$, as claimed.

Every polynomial in $\mathcal{P}_m$ has degree at most 2^m: the largest available degree starts at 1, linear combinations never increase it, and each multiplication at most doubles it. This remains true with stored-value reuse and an arbitrary number of free linear combinations. Consequently,

$$E_m(\delta) \geq r_\delta^{2^m} > 0. \tag{6}$$

Likewise each length-T cubic composition has degree at most 3^T, including degenerate stages, so

$$C_T(\delta) \geq r_\delta^{3^T} > 0. \tag{7}$$

Taking infima in these pointwise inequalities does not require attainment.

3. One cubic step squares the interval error ratio

For $0 < a < 1$, set

$$A = 1 + a + a^2, \qquad M = \frac{2A^{3/2}}{3\sqrt{3}}, \qquad g_a(x) = \frac{x(A - x^2)}{M},$$

and define

$$\phi(a) = \frac{3\sqrt{3}\, a(1+a)}{2(1+a+a^2)^{3/2}}. \tag{8}$$

The polynomial g_a has the allowed form $ux + vx^3$. It maps $[a, 1]$ into $[\phi(a), 1]$. Indeed its unique critical point in that interval is $\sqrt{A/3}$:

$$A - 3a^2 = (1-a)(1+2a) > 0, \qquad 3 - A = (1-a)(a+2) > 0.$$

At this point $x(A - x^2) = M$, while the two endpoint values are both $a(1+a)$. The derivative is positive before the critical point and negative after it. In particular $0 < \phi(a) < 1$.

Lemma 2. For every $0 < a < 1$,

$$\phi(a) \geq \frac{2a}{1+a^2}, \qquad \frac{1-\phi(a)}{1+\phi(a)} \leq \left(\frac{1-a}{1+a}\right)^2. \tag{9}$$

All quantities in the first comparison are positive. After cancelling a and squaring, it is equivalent to

$$27(1+a)^2(1+a^2)^2 - 16(1+a+a^2)^3 \geq 0.$$

The exact identity

$$27(1+a)^2(1+a^2)^2 - 16(1+a+a^2)^3 = (1-a)^2(11a^4 + 28a^3 + 30a^2 + 28a + 11) \tag{10}$$

proves it. Since $v \mapsto (1-v)/(1+v)$ is decreasing for $v > 0$, the first comparison gives the second, using

$$\frac{1 - 2a/(1+a^2)}{1 + 2a/(1+a^2)} = \left(\frac{1-a}{1+a}\right)^2.$$

Starting with $a_0 = \delta$, let $a_{t+1} = \phi(a_t)$. For $T \geq 1$, the composition

$$G_T = g_{a_{T-1}} \circ \cdots \circ g_{a_0}$$

maps $[\delta, 1]$ into $[a_T, 1]$ and is odd. Scale its output by $2/(1 + a_T)$, absorbing that scalar into the two coefficients of the last cubic stage. The resulting allowed length-T composition has error at most

$$\frac{1 - a_T}{1 + a_T} \leq r_\delta^{2^T}.$$

Thus

$$C_T(\delta) \leq r_\delta^{2^T} \qquad (T \geq 1). \tag{11}$$

No additional composition stage is used for the final scaling. Coefficients may depend on T, δ, as permitted by the target. Each stage requires at most two nonscalar multiplications.

4. Strict improvement survives taking infima

Lemma 3. For fixed $\delta \in (0, 1)$, the sequence $C_T(\delta)$ is strictly decreasing in T.

First $C_0(\delta) = 1 - \delta$, and (11) implies

$$C_1(\delta) \leq r_\delta^2 < 1 - \delta = C_0(\delta). \tag{12}$$

For $T \geq 1$, (7) and (11) give $0 < C_T(\delta) < 1$. Take any allowed length-T composition P with error $e < 1$. It is odd and satisfies $P([\delta, 1]) \subset [1 - e, 1 + e]$. Set $a = (1-e)/(1+e)$, so that

$$\frac{P([\delta, 1])}{1 + e} \subset [a, 1].$$

The polynomial in the single variable z,

$$h_e(z) = \frac{2}{1+\phi(a)} g_a\left(\frac{z}{1+e}\right),$$

is one allowed cubic stage. Its composition with P has error at most

$$\frac{1-\phi(a)}{1+\phi(a)} \le \left(\frac{1-a}{1+a}\right)^2 = e^2.$$

Choose a sequence of length-T compositions whose errors tend to $C_T(\delta)$; the errors are eventually less than 1. It follows that

$$C_{T+1}(\delta) \le C_T(\delta)^2 < C_T(\delta). \tag{13}$$

This argument handles potentially unattained infima directly.

5. Proof of the theorem and interpretation

For $m \ge 1$, combine (6) and (11) with $T = m$:

$$C_m(\delta) \le r_\delta^{2^m} \le E_m(\delta). \tag{14}$$

Hence the admissible set in the definition of $T_{\min}$ is nonempty and $T_{\min}(m, \delta) \le m$.

For $m \ge 2$, put $k = \lfloor m/2 \rfloor \ge 1$. Every length-$k$ cubic composition is computed using at most $2k \le m$ multiplications, so

$$E_m(\delta) \le C_k(\delta).$$

If $T < k$, Lemma 3 yields

$$C_T(\delta) > C_k(\delta) \ge E_m(\delta).$$

Thus $T_{\min}(m, \delta) \ge k$, establishing (1).

At $m = 0$, Lemma 1 with $D = 1$ and the linear polynomial $2x/(1+\delta)$ show

$$E_0(\delta) = r_\delta.$$

Every polynomial computable with one multiplication has degree at most two. Its odd part is linear, so the same lower bound proves $E_1(\delta) = r_\delta$. Since $C_0(\delta) = 1 - \delta > r_\delta$ and $C_1(\delta) \le r_\delta^2 < r_\delta$, the asserted two endpoint values follow. Finally, for $m \ge 2$, $\lfloor m/2 \rfloor \ge (m+1)/4$, and the two small cases satisfy (2) as well.

The comparison uses at most $2m$ matrix products for $m \ge 1$, because an allowed cubic stage costs at most two. This is an upper guarantee on multiplication overhead, not an assertion that factor two is optimal or always necessary. The estimate remains uniform when δ tends to either endpoint at any rate with m.

6. Attribution, scope, and verification

The scaled cubic and its endpoint recurrence are due to Chen and Chow (2014), Section 3, equations (3.3)-(3.6), printed pages 9-10; their Theorems 3.1-3.2, printed page 11, analyze its convergence. Polar Express, version 5, Appendix F, equation (17), gives the same cubic and final centering. Its Theorem 3.1, equations (8)-(10), establishes optimality within the fixed-degree composition class, with the Chen-Chow antecedent credited in Section 3.2 and Appendix B. The construction in Section 3 is not a new iteration.

Cheon, Kim and Kim (2020) already establish constant-factor asymptotically optimal fixed-degree composition complexity, including the cubic case. Their relevant locators are Section 2.1, printed page 9; Lemma 3, page 14; Theorem 3, page 15; Table 1, page 17; and Section 3.4, pages 17-18. A short synthesis of their global Lemma 3 and the classical Chebyshev degree bound already gives the uniform constant-factor order for the canonical quantity; the independent source review supplies that synthesis explicitly. No first-discovery or novelty claim is made for that order or the underlying method.

This self-contained note supplies the explicit comparison in (14), hence at most one cubic stage per allowed unrestricted multiplication. The source review did not find that specific comparison in the inspected publications, but that limited search is not a priority certification. The result determines the constant-factor asymptotic order of the canonical MF-02 quantity uniformly in both parameters. The exact minimum, the optimal leading constant, and the stronger source question comparing the two errors at the same multiplication budget remain unanswered. The complete original target and historical references are retained on the canonical page.

Section 2 proves the Chebyshev comparison and Section 3 proves the cubic interval estimate. The independent mathematical review reconstructs the complete argument, including the potentially unattained infima and both endpoint cases. Its exact checker confirms 17 rational polynomial and coefficient-sign checks. These computations supplement the proof. The submission record retains both reviews and their exact source fingerprints. Neither automated-agent review nor these checks constitute external human peer review or formal verification.

References

1. N. Amsel et al., *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, arXiv:2602.05394v3, Section 6.3, Problem 6.5 and equations (21)-(22). Primary source.
2. E. H. Rubensson, E. Jarlebring and G. Lorentzon, *Recursive expansion of the matrix step function using polynomials of degree eight*, arXiv:2606.24701v1, Section 7. Primary source.
3. J. Chen and E. Chow, *A Stable Scaling of Newton-Schulz for Improving the Sign Function Computation of a Hermitian Matrix*, ANL/MCS-P5059-0114, 2014, Section 3, equations (3.3)-(3.6) and Theorems 3.1-3.2. Primary author report.
4. N. Amsel, D. Persson, C. Musco and R. M. Gower, *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm*, arXiv:2505.16932v5, 4 May 2026, Theorem 3.1 and Appendix F, equation (17). Primary source.
5. J. H. Cheon, D. Kim and D. Kim, *Efficient Homomorphic Comparison Methods with Optimal Complexity*, ASIACRYPT 2020, 221-256, Lemma 3, Theorem 3, Table 1 and Section 3.4. DOI; author manuscript.